

Review Exam II

Complex Analysis

Underlined Propositions or Theorems: Proofs May Be Asked for on Exam

Chapter 3.1

Infinite Series

Definition: Convergence

Definition: Absolute Convergence

Proposition 1.1 Absolute Convergence implies Convergence

Examples of series which converge, converge absolutely, converge but do not converge absolutely

Problems about series which converge, converge absolutely, converge but do not converge absolutely

Limit Inferior & Limit Superior

Definition

Examples of $\liminf$, $\limsup$ for which $-\infty \leq \liminf a_n \leq \infty$, $-\infty \leq \limsup a_n \leq \infty$

Problems about $\liminf$, $\limsup$ for which $-\infty \leq \liminf a_n \leq \infty$, $-\infty \leq \limsup a_n \leq \infty$

Power Series

Geometric Series

Theorem 1.1 For $\sum_{n=0}^{\infty} a_n z^n$ let $\frac{1}{R} = \limsup |a_n|^{1/n}$. Then, properties (a), (b) & (c).

Proposition 1.4 Ratio test: For $\sum_{n=0}^{\infty} a_n z^n$, $R = \lim |a_n / a_{n+1}|$, if the limit exists.

Examples of power series with radius of convergence, R , $0 \leq R \leq \infty$

Problems about determining for power series the radius of convergence, R

Propositions on Absolute Convergence

- A) Sum & Cauchy Product of Absolutely Convergent Series is Again Absolutely Convergent
- B) Radius of Convergence for the Sum/Cauchy Product of Power Series is at Least the minimum of the Radii of Convergence of the Summands/Multiplicands
- C) Absolutely Convergent Series can be Rearranged

Assume the following convention throughout the remainder of these notes, unless otherwise specified:

G = region (open, connected subset of $\mathbb{C}$); $f, g: G \rightarrow \mathbb{C}$

Chapter 3.2

Differentiability and Analyticity

Definition of differentiability of f at a point in G

Definition of differentiability of f on G

Definition of continuous differentiability of f on G

Definition of analyticity of f at a point in G

Definition of analyticity of f on G

Conway: analyticity of f on $G \equiv$ continuous differentiability of f on G

Definition: $A(G)$

Definition: set of *entire* functions = $A(\mathbb{C})$

Proposition If f is differentiable at $a \in G$, then f is continuous at a

Proposition Let $f, g \in A(G)$ and let $a \in \mathbb{C}$, then $af, f+g, f-g, fg \in A(G)$ and $f/g \in A(G \setminus \overline{Z_g})$

Proposition Let $a \in \mathbb{C}$ and let $f(z) \equiv a, g(z) \equiv z$. Then, $f, g \in A(\mathbb{C})$.

Corollary Let f be a polynomial and let $g = \frac{p}{q}$ be a rational function. Then, $f \in A(\mathbb{C})$ and $g \in A(\mathbb{C} \setminus \overline{Z_q})$.

Proposition 2.5 Let $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ have radius of convergence $R = R_f > 0$

a) Then, $g(z) = \sum_{n=1}^{\infty} na_n (z - z_0)^{n-1}$ has radius of convergence $R_g \geq R_f$ (Actually equal)

b) $f \in A(B(z_0, R))$ and $f' = g$

c) f is infinitely differentiable on $B(z_0, R)$ and $f^{(k)}(z) = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} a_n (z - z_0)^{n-k}$

d) $a_n = \frac{1}{n!} f^{(n)}(z_0)$

Proposition 2.10 Let $f \in A(G)$. If $f' \equiv 0$ on G , then f is constant on G .

Complex Exponential Function

Definition e^z

Differentiability $(e^z)' = e^z$

Properties $e^{z+w} = e^z e^w$, $e^z \neq 0$, $|e^z| = e^{\operatorname{Re} z}$, $e^{it} = \operatorname{cis} t$ (for t real), Periodicity,
 $e^z = e^{x+iy} = e^x e^{iy} \Rightarrow |e^z| = e^x$, $\arg e^z = y$, Euler's Equation,
 $e^z = 1$ iff $z = 2\pi ki$ for some integer k

Complex Trig Functions

Definition $\cos(z)$, $\sin(z)$

Differentiability $(\cos z)' = -\sin z$, $(\sin z)' = \cos z$

Properties Pythagorean Theorem, Addition Formulas, Representation Formulas between Complex Exponential and Complex Trig, Periodicity, Zeros

Other Trig Functions $\tan z$, $\sec z$, $\cot z$, $\csc z$

Complex Hyperbolic Trig Functions

Definition $\cosh(z)$, $\sinh(z)$

Differentiability $(\cosh z)' = \sinh z$, $(\sinh z)' = \cosh z$

Properties Pythagorean Theorem, Addition Formulas, Representation Formulas between Complex Exponential and Complex Hyperbolic Trig, Representation Formulas between Complex Trig and Complex Hyperbolic Trig

Complex Logarithm

Definition of a *branch of $\log z$* on G

Implication that $0 \notin G$

Proposition 2.19 Totality of branches of $\log z$ on a region G

Definition of principal branch of $\log z$ on $\mathbb{C} \setminus (-\infty, 0]$

$$\log z = \log |z| + i \operatorname{Arg} z \text{ where } -\pi < \operatorname{Arg} z < \pi$$

Proposition 2.20 Differentiability of a continuous inverse

Differentiability $(\log z)' = \frac{1}{z}$ (for any branch of $\log z$)

Complex Powers

Definition $z^b = \exp(b \log z)$ via principal branch of $\log z$

Differentiability $(z^b)' = b z^{b-1}$

Cauchy-Riemann Equations

Let $f = u + iv$. Then, $f \in A(G)$ iff the partials u_x, u_y, v_x, v_y exist and are continuous on G and

$$\begin{cases} u_x = v_y \\ v_x = -u_y \end{cases} \text{ on } G.$$

Proposition Let $f = u + iv$. Let $f \in A(G)$. Then, u and v are harmonic on G .

Definition: Harmonic Conjugate

Proposition $\log x^2 + y^2$ is harmonic on $\mathbb{C} \setminus \{0\}$ but does not have a harmonic conjugate on $\mathbb{C} \setminus \{0\}$

Proposition 2.30 Let G be a disk or the entire complex plane. If u is harmonic on G , then u has a harmonic conjugate on G

Mapping Properties of Standard Functions

$e^z : G \rightarrow \mathbb{C}$ where G is a rectangle of the form $G = \{z = x + iy : a < x < b, c < y < d\}$

Ex 1: $G = \{z : |\operatorname{Im} z| < p\}$

Ex 3: $G = \{z : 0 < \operatorname{Im} z < p\}$

Ex 2: $G = \{z : |\operatorname{Im} z| < p/2\}$

Ex 4: $G = \{z : 0 < \operatorname{Im} z < p/2\}$

$z^n : G \rightarrow \mathbb{C}$ where G is a sector of the form $G = \{z = re^{iq} : a < r < b, c < q < d\}$

Ex 1: $n = 2, 3, 4, \dots$ and G is the first quadrant

Ex 2: $n = 2, 3, 4, \dots$ and G is the upper half-plane

Ex 3: $n = 2, 3, 4, \dots$ and G is the right half-plane

Ex 4: $n = 2, 3, 4, \dots$ and G is $B(0, R) \cap$ first quadrant

$\sin z : G \rightarrow \mathbb{C}$ where G is the base half-strip for $\sin z$, i.e., $G = \{z : |\operatorname{Re} z| < p/2 \text{ and } \operatorname{Im} z > 0\}$

$\log z : G \rightarrow \mathbb{C}$ where G is a subset of $\mathbb{C} \setminus (-\infty, 0]$, specifically where G is a sector of the form

$G = \{z = re^{iq} : a < r < b, c < q < d\}$

Ex 1: G is the first quadrant

Ex 4: G is $B(0, R) \cap$ first quadrant

Ex 2: G is the upper half-plane

Ex 5: G is $\mathbb{C} \setminus (-\infty, 0]$

Ex 3: G is the right half-plane

$z^{1/n} : G \rightarrow \mathbb{C}$ where G is a sector of the form $G = \{z = re^{iq} : a < r < b, c < q < d\}$

Ex 1: $n = 2, 3, 4, \dots$ and G is the first quadrant

Ex 2: $n = 2, 3, 4, \dots$ and G is the upper half-plane

Ex 3: $n = 2, 3, 4, \dots$ and G is the right half-plane

Ex 4: $n = 2, 3, 4, \dots$ and G is $B(0, R) \cap$ first quadrant

Ex 5: G is $\mathbb{C} \setminus (-\infty, 0]$

Conformality

Definition: path, smooth path, piece-wise smooth path

Definition: tangent direction to smooth path g at point $z = g(t)$ where $g'(t) \neq 0$

Definition: angle between two smooth paths at a point $z_0 = g_1(t_1) = g_2(t_2)$ where $g_1'(t_1) \neq 0, g_2'(t_2) \neq 0$

Theorem 3.4 Let $f \in A(G)$ and let $z_0 \in G$. If $f'(z_0) \neq 0$, then f “preserves angles” for any smooth paths $g_1, g_2 \in G$ such that $z_0 = g_1(t_1) = g_2(t_2)$ and $g_1'(t_1) \neq 0, g_2'(t_2) \neq 0$

Definition: Conformality

Chapter 3.3

Bi-Linear, Linear Fractional, Moebius Transformations

Definition $S(z) = \frac{az + b}{cz + d}, ad - bc \neq 0$

Differentiability $S'(z) = \frac{ad - bc}{(cz + d)^2}$

Properties $S : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty, S$ is 1-1, S is onto, pole of S is $-d/c$, zero of S is $-b/a$

$T(z) = S^{-1}(z) = \frac{-dz + b}{cz - a} = \frac{dz - b}{-ca + a}, S$ conformal on $\mathbb{C} \setminus \{-d/c\}$, Fixed Points,

Uniqueness

Special Cases: Translations, Dilations, Rotations, Inversion

Every Bi-Linear Transformation can be written as a composition of Translations, Dilations, Rotations, and the Inversion

Mapping Properties of Special Cases: “Circles” to “Circles”

Geometry of Images